

Four-input ordered coefficients and native Heisenberg collisions

Abstract

The two disjoint adjacent collisions of four ordered inputs define a square of coefficient forms. Weighted integration on this square supplies the double-pair component of a native Heisenberg comparison. Its boundary identity determines the remaining defects on the triple, double-pair, and small diagonals. An explicit complex of translated current modules absorbs these defects at the original nonzero level. The construction gives a permutation-equivariant cochain map for every global four-input coefficient. The cohomology of the added complex and the resulting relative cohomology are described by Weyl-module presentations. Ordered three-plus-one cuts agree with the ternary comparison on their factorization domains. A torsion calculation obstructs the corresponding strict two-plus-two extension.

TRANSLATED ORDERED STATES AND NATIVE HEISENBERG COLLISIONS

1.1 THE TRANSLATION OF AN ORDERED PRODUCT

Fix $k \in \mathbb{C}^\times$ and $X = \mathbb{A}_{\mathbb{C}}^1$. All position functions are algebraic, and every differential operator has finite order. No completion or additional state differential is used. For two positions put $t = z_1 - z_2$ and $y = z_2$. The coefficient algebra is

$$\begin{aligned} C_2^o &= \{f \in \mathbb{C}[y, t, t^{-1}][u] : f(o), f(1) \in \mathbb{C}[y, t]\}, \\ C_2^1 &= \mathbb{C}[y, t, t^{-1}][u]du, \quad d_C f = \partial_u f du. \end{aligned}$$

The zero-jet ordered product is $a_1 a_2 = \omega_{12} b_{12}$, where $\omega_{12} = du/t$, $|a_i| = o$, and $|b_{12}| = -1$. Translations of the inputs must also differentiate ω_{12} . For example, a translation derivation L_1 with $L_1 z_1 = 1$ satisfies

$$(L_1 a_1) a_2 = -\frac{du}{t^2} b_{12} + \frac{du}{t} L_1 b_{12}. \quad (1.1)$$

The first summand is required even when b_{12} has a prescribed translation.

For $n = 2, 3$, write $R_n = \mathbb{C}[z_1, \dots, z_n]$. Here is a common definition of the coefficient algebra C_n . Let O_n be the total orders of the labels. For nonempty $A \subset O_n$, let R_A invert $z_i - z_j$ when the relative order of i, j varies in A . Let

$$\Omega_A = \mathbb{C}[u_\sigma, du_\sigma \mid \sigma \in A] / (\sum_A u_\sigma - 1, \sum_A du_\sigma).$$

The odd generators du_σ form an exterior algebra. A member of C_n is a family in $R_A \otimes \Omega_A$ compatible with every face restriction and coefficient localization. The differential sends u_σ to du_σ and fixes positions. Set

$$q_{ij} = \sum_{i \prec_\sigma j} u_\sigma, \quad \omega_{ij} = \frac{dq_{ij}}{z_i - z_j}.$$

On a face where the denominator is unavailable, $dq_{ij} = o$ and that component is zero. The position derivations $\partial_i = \partial_{z_i}$ preserve C_n and commute with d_C . Indeed, they preserve every localization and every face condition.

Adjoin closed symbols $a_i^{(r)}$ and $b_{ij}^{(r,s)}$, for $r, s \geq o$, of degrees o and -1 , respectively. Coefficients act graded centrally. Define the only nonzero nonunit products by

$$a_i^{(r)} a_j^{(s)} = \sum_{p=o}^r \sum_{q=o}^s \binom{r}{p} \binom{s}{q} (\partial_i^p \partial_j^q \omega_{ij}) b_{ij}^{(r-p, s-q)}, \quad i \neq j. \quad (1.2)$$

Products involving a b and another nonunit symbol are zero. Products of two a_i with the same label are zero. Let L_i act by ∂_i on coefficients. It increases the i -label jet index on a symbol and kills symbols without label i . The operators L_i commute. Total translation of a two-label symbol is $L_i + L_j$.

PROPOSITION 1.1.1. Formula (1.2) defines an associative differential graded algebra over C_n . Each L_i is a degree-zero derivation commuting with the differential. The untranslated symbols have exactly the product $a_i^{(o)} a_j^{(o)} = \omega_{ij} b_{ij}^{(o,o)}$.

Proof. Every product of three nonunit symbols vanishes under either parenthesization. This proves associativity. The coefficients in (1.2) are closed, since $d_C \omega_{ij} = 0$ and d_C commutes with position derivatives. Thus the differential obeys the graded Leibniz identity. Applying L_i differentiates the coefficient or increases the first b index. Pascal's identity combines these terms into (1.2) with r replaced by $r + 1$. The same calculation with $s + 1$ proves the L_j identity. Every other L_l fixes the kernel and both input labels. This proves the derivation assertion on all products. $\square$

1.2 THE RIGHT DIFFERENTIAL MODULE OF ORDERED STATES

Suspension lowers degree by one and satisfies $s(ca) = (-1)^{|c|} c sa$. The bar convention is $b_2(sa \otimes sb) = (-1)^{|a|} s(ab)$. For $n = 2$, the untranslated sector has symbols $e = [a_1|a_2]$ and $h = [b_{12}]$, both of degree -2 . Its differential is $be = -\omega_{12}h$ and $bh = 0$. For $n = 3$ and $\sigma = ijk$, use

$$e_\sigma = [a_i|a_j|a_k], \quad m_\sigma = [b_{ij}|a_k], \quad n_\sigma = [a_i|b_{jk}].$$

These eighteen symbols have degree -3 . Their free C_3 -module O_3 has differential

$$be_\sigma = -\omega_{ij}m_\sigma - \omega_{jk}n_\sigma, \quad bm_\sigma = bn_\sigma = 0, \quad b(cv) = d_C c v + (-1)^{|c|} c bv. \quad (1.3)$$

Denote the binary complex by O_2 .

Let P_n be the same bar sector with all translated symbols from (1.2). For a multi-index $\alpha \in \mathbb{N}^n$, write v_α for the symbol whose input-label jets are α . Each P_n is the direct sum of $C_n v_\alpha$ over its untranslated symbols and finite jet indices. On these words,

$$L_i(cv_\alpha) = (\partial_i c)v_\alpha + cv_{\alpha+e_i}.$$

For the three-letter symbol the bar differential is explicitly

$$bv_\alpha = L^\alpha bv_0, \quad b(e_\sigma)_\alpha = - \sum_{\gamma \leq \alpha} \binom{\alpha}{\gamma} \left((\partial^\gamma \omega_{ij})(m_\sigma)_{\alpha-\gamma} + (\partial^\gamma \omega_{jk})(n_\sigma)_{\alpha-\gamma} \right). \quad (1.4)$$

Mixed words have zero symbol differential. The binary formula omits the second summand. The graded coefficient rule in (1.3) remains in force.

Let D_n be the Weyl algebra of R_n , with $[\partial_i, z_j] = \delta_{ij}$. The right action on P_n is

$$w \partial_i = -L_i w, \quad wf = fw \quad (f \in R_n).$$

The identity $(w \partial_i)f - (wf)\partial_i = w(\partial_i f)$ verifies the Weyl relation. Flatness of the L_i verifies all derivative commutators. Equation (1.4) gives $bL_i = L_i b$, so P_n is a complex of right D_n -modules. The untranslated inclusion $O_n \hookrightarrow P_n$ is a cochain map over R_n .

PROPOSITION 1.2.1. There is an isomorphism of right differential-module complexes

$$O_n \otimes_{R_n} D_n \longrightarrow P_n, \quad x \otimes A \longmapsto xA. \quad (1.5)$$

Consequently,

$$H^r(P_n) = H^r(O_n) \otimes_{R_n} D_n. \quad (1.6)$$

This is not an identification of P_n with O_n .

Proof. Balancing follows from the stated regular-function action. The ordered monomials ∂^α form a free left R_n -basis of D_n . For $x = cv_0$, the image of $x \otimes \partial^\alpha$ is $(-1)^{|\alpha|} cv_\alpha$ plus terms with smaller jet order. This triangular formula gives an inverse on every finite sum. Right linearity is built into the definition. The chain identity follows from $bL_i = L_ib$. Finally, left R_n -freeness of D_n makes tensoring exact, proving (1.6). $\square$

The products (1.2) specify what the induction means for translated states. Keeping the same formula $a_i^{(r)} a_j^{(s)} = \omega_{ij} b_{ij}^{(r,s)}$ for every jet would contradict (1.1).

1.3 THE UNEXTENDED HEISENBERG TARGET

The Heisenberg states, vacuum, current, and translation are

$$V_k = \mathbb{C}[x_1, x_2, \dots], \quad \mathbf{I} = \mathbf{I}, \quad J = x_1, \quad T = \sum_{r \geq 1} r x_{r+1} \partial_{x_r}.$$

Every state is a finite polynomial, is even, and has degree zero. The modes are $J_{-r} = x_r$, $J_r = kr \partial_{x_r}$ for $r \geq 1$, and $J_0 = 0$. The field of a monomial is the normally ordered product of the divided current derivatives. Normal ordering puts multiplication operators before derivative operators. Commuting these operators gives the finite Wick pairing rule. For factors x_r and x_l , a contraction is

$$C_{rl}(t) = \frac{k(-1)^{r-1}(r+l-1)!}{(r-1)!(l-1)!} t^{-r-l}.$$

Sum over all partial pairings of distinct input factors. Each pair contributes its contraction, and the unpaired factors remain normally ordered. The rule follows by applying $AB = BA + [A, B]$ until every multiplication operator precedes every derivative operator. Each contracted factor is removed, so every fixed monomial has finitely many pairings. In particular,

$$p(t) = Y(J, t)J = \frac{k}{t^2} \mathbf{I} + (e^{tT} J)J, \quad Y(\mathbf{I}, t)a = a, \quad T^r J = r! x_{r+1}.$$

The right differential module $\mathcal{H}_k = \omega_X \otimes V_k$ has action $(f a dz) \partial_z = -(f' a + f T a) dz$. For $t = z - w$ and $y = w$, its chiral operation is

$$\mu(f(z, w)(a dz \boxtimes b dw)) = \text{pp}_t(f(y + t, y)Y(a, t)b), \quad \text{pp}_t \frac{a(y)}{t^{r+1}} = \frac{a(y)dy}{r!} \delta \partial_t^r. \quad (1.7)$$

The orientation is $dz \wedge dw = dt \wedge dy$. Multiplication by a Laurent coefficient occurs before taking the principal part. An existing principal-part class only admits multiplication by regular functions. The normal action is $-\partial_t$ on Laurent representatives. In particular, $(\partial \partial_t^r)t = r \partial \partial_t^{r-1}$.

Translation of fields follows directly from the mode formulas: $Y(Ta, t) = \partial_t Y(a, t)$ and $TY(a, t)b = \partial_t Y(a, t)b + Y(a, t)Tb$. These two equations make (1.7) a right differential-module map. The contraction relation $C_{lr}(-t) = C_{rl}(t)$ gives vertex skew symmetry by interchanging the two inputs in the Wick sum. Normal coordinate transposition acts as $p(t, y) \mapsto -\text{pp}_t(e^{tT} p(-t, y + t))$. The density minus sign therefore gives skew symmetry for the chiral operation.

For three monomial states the Wick expression belongs to $V_k[[u, v]][u^{-1}, v^{-1}, (u - v)^{-1}]$. Its two radial expansions are the two ordered products. Pairing the first two inputs first gives its expansion along $u = v$, hence the iterate. The following residue identity gives their difference:

$$\begin{aligned} \text{Res}_v \text{Res}_t G(v + t, v) dt dv &= \text{Res}_u \text{Res}_v i_{u,v} G(u, v) dv du \\ &\quad - \text{Res}_v \text{Res}_u i_{v,u} G(u, v) du dv. \end{aligned}$$

Here $i_{u,v}$ expands in nonnegative powers of v/u . For a rational numerator the identity is the partial-fraction identity for the poles $u = 0$ and $u = v$. For a power-series numerator, each residue uses finitely many coefficients, reducing to that case. Apply the identity after multiplication by every polynomial testing the two normal variables. This proves chiral Jacobi coefficient by coefficient, including input translations and finite coefficient poles.

For a partition π of $\{1, \dots, n\}$, let Δ_π identify its blocks and let j_π exclude collisions between distinct blocks. The native complex is

$$E_n = \bigoplus_{\pi} (\Delta_\pi)_* (j_\pi)_* j_\pi^* (\boxtimes_{B \in \pi} \mathcal{H}_k[1]). \quad (1.8)$$

The closed direct image uses the right transfer module with all finite normal derivatives. There is no additional degree shift. Formula (1.8) is the Chevalley–Cousin carrier of [1], §3.4.II, equation (3.4.II.1). Its partition summand has degree $-\#\pi$. For even states the collision differential is

$$Q[a|b] = [\mu(a, b)], \quad Q[a|b|c] = [\mu_{12}(a, b)|c] - [\mu_{13}(a, c)|b] + [\mu_{23}(b, c)|a]. \quad (1.9)$$

Every map includes transverse localization and the density permutation. Swapping suspended even blocks changes the sign. The preceding Jacobi identity gives $Q^2 = 0$.

1.4 THE BINARY COMPARISON AND ITS SOURCE TERM

For an actual edge coefficient define

$$\begin{aligned} A(g) &= \int_0^1 g(u) du, & M(g) &= \int_0^1 u g(u) du, \\ q_g &= \text{pp}_t(t p(t) A(g)), & v_g &= \text{pp}_t(p(t) M(g)). \end{aligned} \quad (1.10)$$

Characteristic zero makes these polynomial integrals exact. Every normal coefficient is regular in the remaining block positions. Integration by parts gives

$$v_{\partial_u f} + q_{f|t} = \text{pp}_t(p(t) f(1)), \quad q_{\partial_u f} = k \text{pp}_t \frac{f(1) - f(0)}{t} \mathbf{1}. \quad (1.11)$$

The second identity uses endpoint regularity and the factor t in q_g .

Define a degree-zero R_2 -linear map $B_2 : O_2 \rightarrow E_2$ by

$$B_2(fe) = f(1)[J_1|J_2], \quad B_2(gdu e) = [v_g], \quad B_2(fh) = 0, \quad B_2(gdu h) = -[q_g].$$

Let N_2 be the right differential module generated by $\partial_{12} \mathbf{1}$ in degree -1 . It is closed under Q , and

$$N_2 \cong D_2 / (t D_2 + \partial_y D_2). \quad (1.12)$$

Its independent forms are $f(y) \mathbf{1} \partial_{12} \partial_t^r$, $r \geq 0$.

Extend B_2 to P_2 by right differential linearity through (1.5). This is an actual map on translated states. For any base coefficient c and word v , its recursion is

$$B_n(cv_{\alpha+e_i}) = -B_n(cv_\alpha) \partial_i - B_n((\partial_i c) v_\alpha). \quad (1.13)$$

For an unmerged zero-form it sends the i -th jet to T on precisely the i -th native current. The identity holds for every jet order, because evaluation commutes with position differentiation.

PROPOSITION 1.4.1. The binary defect $\beta_2 = QB_2 - B_2b$ takes values in N_2 . On untranslated inputs its only nonzero component is

$$\beta_2(fh) = k \text{ pp}_t \frac{f(\mathbf{i}) - f(\mathbf{o})}{t} \mathbf{I}.$$

The complex

$$\tilde{P}_2 = P_2 \oplus N_2, \quad \tilde{b}(x, n) = (bx, \beta_2 x)$$

admits the native cochain map $F_2(x, n) = B_2x + n$. Among added submodules of the target for this fixed B_2 and splitting, N_2 is minimal.

Proof. For fe , the first identity in (1.11) gives zero defect. The target differential of a one-form image is zero by degree. For fh , the defect is $q_{\partial_u}f$, giving the displayed formula. Right differential linearity covers all jets. Since $Q^2 = b^2 = \mathbf{o}$, one has $Q\beta_2 + \beta_2b = \mathbf{o}$. The target differential is zero on N_2 , hence $\beta_2b = \mathbf{o}$. This proves $\tilde{b}^2 = \mathbf{o}$ and $QF_2 = F_2\tilde{b}$. Taking $f = u$ gives $k\delta_{12}\mathbf{I}$. Its right differential closure is all of (1.12), proving minimality. $\square$

The binary cohomology change is explicit in its lowest degree. Endpoint integration gives

$$H^{-2}(O_2) = R_2z \oplus R_2h, \quad z = te + uh.$$

Indeed, a cycle $fe + gh$ has $f \in R_2$ and $\partial_u g = f/t$. Regular endpoints force $t \mid f$, and subtracting a multiple of z leaves a constant multiple of h . Its connecting map sends z to $k\delta_{12}\mathbf{I}$ and h to zero. Therefore

$$H^{-2}(\tilde{P}_2) = z(tD_2 + \partial_y D_2) \oplus hD_2 \subset H^{-2}(O_2) \otimes_{R_2} D_2.$$

All other cohomology groups of the projection $\tilde{P}_2 \rightarrow P_2$ are isomorphic. Surjectivity of the displayed connecting map proves this assertion from the long exact sequence.

1.5 ALL THREE COLLISION DIAGONALS

For each word ijk , restrict coefficients to its edges $\{ijk, jik\}$ and $\{ijk, ikj\}$. Their coordinates are $u_L = q_{ij}$ and $u_R = q_{jk}$. Both coordinates equal one at the word vertex. For $c \in C_3^1$, write $c|_L = g_L du_L$ and $c|_R = g_R du_R$. Use (1.10) in each edge normal coordinate. Define $B_3 : O_3 \rightarrow E_3$ by

$$\begin{aligned} B_3(fe_{ijk}) &= f(ijk)[J_i|J_j|J_k], \\ B_3(ce_{ijk}) &= [v_{g_L}|J_k] - [J_i|v_{g_R}], \\ B_3(fm_\sigma) &= B_3(fn_\sigma) = \mathbf{o}, \\ B_3(cm_\sigma) &= -[q_{g_L}|J_k], \\ B_3(cn_\sigma) &= +[J_i|q_{g_R}]. \end{aligned} \tag{1.14}$$

Here f has degree zero and c has degree one. Every component of coefficient degree at least two has zero image. The mixed formulas are the binary images with the suspended spectator and its tensor sign. Extend to P_3 by (1.13).

For a finite inner principal part w , define $R_L(w) = Q[w|J]$ and $R_R(w) = Q[J|w]$. These operations retain every inner normal derivative. They take the outer principal part only after transverse localization. Put $\Delta_L f = f(ijk) - f(jik)$ and $\Delta_R f = f(ijk) - f(ikj)$.

LEMMA 1.5.1. The defect $\beta_3 = QB_3 - B_3b$ on untranslated words is

$$\begin{aligned}
\beta_3(fe_{ijk}) &= -[\mu_{ik}(f(ijk)(J_i, J_k))|J_j], \\
\beta_3(ce_{ijk}) &= R_L(v_{g_L}) - R_R(v_{g_R}), \\
\beta_3(fm_\sigma) &= k[\text{pp}_{t_{ij}}(\Delta_L f / t_{ij})\mathbf{1}|J_k], \\
\beta_3(fn_\sigma) &= -k[J_i|\text{pp}_{t_{jk}}(\Delta_R f / t_{jk})\mathbf{1}], \\
\beta_3(cm_\sigma) &= -R_L(q_{g_L}), \\
\beta_3(cn_\sigma) &= +R_R(q_{g_R}).
\end{aligned} \tag{1.15}$$

It vanishes on all higher coefficient degrees.

Proof. For a zero-form unmerged word, the two adjacent components cancel by the first identity in (1.11). The nonadjacent component has the middle unshuffle sign in (1.9). For a one-form unmerged word, all terms in B_3b restrict to two-forms on an edge and vanish. Its two outer collisions give the second line. On mixed zero-forms use the second identity in (1.11). On mixed one-forms the source differential again has zero edge image. The target differential gives the last two lines with the signs in (1.14). Higher coefficients vanish for the same edge-dimension reason. $\square$

1.6 THE MINIMAL TERNARY ADDED MODULE

Let $U_J = \text{span}_{\mathbb{C}}\{T^r J : r \geq 0\}$ with its stated translation. For each unordered pair ij with spectator l , let M_{ij} be the regular-coefficient right differential closure of $[\mathbf{1}\delta_{ij}|J_l]$ inside the $ij|l$ target summand. The pair block is listed first, and $t_{ij} = z_i - z_j$ uses $i < j$. Let S_3 be the right differential closure of $[J\delta_{123}]$ in the small-diagonal summand. Then

$$\begin{aligned}
M_{ij} &\cong D_3 / (t_{ij}D_3 + (\partial_i + \partial_j)D_3), \\
S_3 &\cong D_3 / (t_{12}D_3 + t_{23}D_3).
\end{aligned}$$

The vacuum fixes the first tangent relation. The spectator derivatives generate the independent states $T^r J$. For S_3 , the total tangent derivative generates those same independent states, so there is no tangent annihilator. Normal ordering in the Weyl algebra gives independent finite transfer forms in both presentations. This proves completeness of their relations.

Put $M_3 = M_{12} \oplus M_{13} \oplus M_{23}$ in degree -2 and put S_3 in degree -1 . Their direct sum $N_3 = M_3 \oplus S_3$ has zero differential. Indeed, a pair term has vacuum in its merged block and regular outer coefficients. Its remaining principal part is zero by $Y(\mathbf{1}, s)J = J$. Right differential linearity preserves this vanishing. The small-diagonal term has no further collision.

THEOREM 1.6.1. The defect β_3 has values in N_3 . The right differential-module complex

$$\bar{P}_3 = P_3 \oplus N_3, \quad \bar{b}(x, n) = (bx, \beta_3 x) \tag{1.16}$$

admits the native cochain map $F_3(x, n) = B_3 x + n$ on all of X^3 . It is defined on every ordered word, every global compatible-family coefficient, and every finite input translation. For this prescribed B_3 , and an added module included in the target with the displayed splitting, N_3 is minimal.

Proof. The zero-form unmerged output is a regular coefficient times $k\delta_{ik}\partial_{t_{ik}}$ with spectator J_j . The mixed zero-form outputs have vacuum in the pair block by (1.15). Thus all degree-2 outputs lie in M_3 .

Every v_g or q_g has vacuum states or quadratic current states, including their translations. Its normal coefficients are regular in the outer variable. The outer collision of a vacuum term is zero. In the Wick expansion of a quadratic current state against one current, a singular term uses exactly one contraction. It leaves exactly one translated current. Therefore every degree-1 output lies in S_3 . The same conclusion holds after all right derivatives.

The identity $Q\beta_3 + \beta_3 b = 0$ follows by expanding the defect. Since $QN_3 = 0$, it gives $\beta_3 b = 0$. Thus (1.16) is a complex, and direct substitution gives $QF_3 = F_3 \tilde{b}$.

For any pair choose a mixed word containing it and the coefficient q_{ij} . The corresponding defect is, up to its displayed sign, $k[\mathbf{1}\delta_{ij}|J_l]$. Its right differential closure is M_{ij} . It remains to show that S_3 is forced. Take $t = z_1 - z_2$, $s = z_2 - z_3$, $y = z_3$, and the global closed coefficient

$$\eta_{12} = \frac{dq_{12}}{t^2}. \quad (1.17)$$

On a face with constant relative order, its numerator is zero. On every other face the coefficient ring inverts t . Thus this is an actual member of C_3^1 . For $m = m_{12|3}$, its left edge coefficient is $g = 1/t^2$, and

$$q_g = \frac{k}{2}\mathbf{1}\delta_{12}\partial_t^2 + J^2\delta_{12}, \quad \beta_3(\eta_{12}m) = -2k(J\delta_{123}\partial_s + (TJ)\delta_{123}). \quad (1.18)$$

For the second formula, the inner vacuum term has zero outer output and $\text{pp}_s Y(J^2, s)J = 2k(J/s^2 + TJ/s)$. The orientation is $dz_1 \wedge dz_2 \wedge dz_3 = dt \wedge ds \wedge dy$. Multiplication by s gives $-2kJ\delta_{123}$. This nonzero element generates S_3 under right differential operators. Every added target submodule that absorbs the defect must therefore contain N_3 . $\square$

The factor t in q_g is decisive. The coefficient dq_{12}/t gives only $k\mathbf{1}\delta_{12}\partial_t$ and has zero outer collision. The first nonzero mixed outer output is (1.17), which has one additional inner pole. The new small-diagonal source term is exactly (1.18).

1.7 COHOMOLOGY RELATIVE TO THE ORIGINAL ORDERED COMPLEX

All cohomology in this section is cohomology of algebraic module complexes. Write $H_D^r = H^r(O_3) \otimes_{R_3} D_3$. The added module gives the short exact sequence

$$0 \longrightarrow N_3 \longrightarrow \tilde{P}_3 \longrightarrow P_3 \longrightarrow 0.$$

Its connecting maps are the induced defect maps $\delta_{-3} : H_D^{-3} \rightarrow M_3$ and $\delta_{-2} : H_D^{-2} \rightarrow S_3$. They are defined explicitly by (1.15) and right differential linearity.

For each $\sigma = ijk$, put $p_\sigma = t_{ij}t_{jk}$ and

$$z_\sigma = p_\sigma e_\sigma + t_{jk}q_{ij}m_\sigma + t_{ij}q_{jk}n_\sigma.$$

These are cycles, and

$$H^{-3}(O_3) = \bigoplus_{\sigma} (R_3 z_\sigma \oplus R_3 m_\sigma \oplus R_3 n_\sigma). \quad (1.19)$$

To prove completeness, a closed unmerged coefficient is constant on the full simplex. Its vertex values force it to belong to R_3 . For a cycle $f e_\sigma + g m_\sigma + h n_\sigma$, integration on its two actual edges gives $t_{ij} \mid f$

and $t_{jk} \mid f$. The two relatively prime linear polynomials give $p_\sigma \mid f$. Subtracting the corresponding multiple of z_σ leaves constant mixed coefficients. There are no incoming boundaries in degree -3 . This proves (1.19).

In fixed label order set

$$c = k[\mathbf{1}\delta_{12}\delta_{12}|J_3] - k[\mathbf{1}\delta_{13}\delta_{13}|J_2] + k[\mathbf{1}\delta_{23}\delta_{23}|J_1] \in M_3.$$

Every coordinate permutation here uses (1.9) and its density identification. If ε_σ is the sign of the permutation σ , then

$$\delta_{-3}(z_\sigma) = \varepsilon_\sigma p_\sigma c, \quad \delta_{-3}(m_\sigma) = \delta_{-3}(n_\sigma) = 0.$$

Indeed, $B_3 z_\sigma = p_\sigma [J_i | J_j | J_k]$ and $b z_\sigma = 0$. Apply Q and use its regular-function linearity.

This also gives a finite right-module presentation for the cohomology contribution from the pair diagonals. Put $t = z_1 - z_3$ and $s = z_2 - z_3$ in the next formula only. The six polynomials p_σ generate (t^2, ts, s^2) in R_3 . The relations $p_\sigma c$ therefore generate precisely

$$K = c t^2 D_3 + c t s D_3 + c s^2 D_3 \subset M_3. \quad (1.20)$$

The right differential closure removes this quotient completely.

LEMMA 1.7.1. One has $K = M_3$.

Proof. Let $g_{12} = [\mathbf{1}\delta_{12}|J_3]$ and temporarily put $x = t_{12}, s = t_{23}$. The element $c t_{13}^2 t_{23}^2$ belongs to K by (1.20). Its 13 and 23 components vanish, since a double-pole class is killed by its squared normal coordinate. On the 12 component, $t_{13} = x + s$ and normal multiplication gives

$$(c t_{13}^2 t_{23}^2)x = k g_{12} s^4.$$

Thus the class $\bar{g}_{12}$ in M_3/K satisfies $\bar{g}_{12} s^4 = 0$. It also satisfies $\bar{g}_{12} P = 0$, where $P = \partial_1 + \partial_2$. The Weyl relation is $[P, s] = 1$. If $\bar{g}_{12} s^r = 0$ for $r > 0$, then

$$0 = (\bar{g}_{12} s^r) P = (\bar{g}_{12} P) s^r - r \bar{g}_{12} s^{r-1} = -r \bar{g}_{12} s^{r-1}.$$

Descending from $r = 4$ gives $\bar{g}_{12} = 0$. The two relabelings give the same conclusion for g_{13} and g_{23} . These three elements generate M_3 , proving the assertion. $\square$

THEOREM 1.7.2. Both connecting maps $\delta_{-3} : H_D^{-3} \rightarrow M_3$ and $\delta_{-2} : H_D^{-2} \rightarrow S_3$ are surjective. The entire cohomology change from P_3 to $\tilde{P}_3$ is described by

$$\begin{aligned} 0 &\longrightarrow H^{-3}(\tilde{P}_3) \longrightarrow H_D^{-3} \xrightarrow{\delta_{-3}} M_3 \longrightarrow 0, \\ 0 &\longrightarrow H^{-2}(\tilde{P}_3) \longrightarrow H_D^{-2} \xrightarrow{\delta_{-2}} S_3 \longrightarrow 0, \\ H^r(\tilde{P}_3) &\cong H_D^r \quad (r \neq -3, -2). \end{aligned} \quad (1.21)$$

The first kernel has the explicit equation $\sum_\sigma \varepsilon_\sigma p_\sigma c A_\sigma = 0$ for the coefficients $A_\sigma \in D_3$ of z_σ . Every mixed basis coefficient is unrestricted there.

Proof. The element $\eta_{12}m_{12|3}$ is closed because $d_C\eta_{12} = 0$ and $bm_{12|3} = 0$. Its connecting image is (1.18). Multiplication by s gives a nonzero scalar times the cyclic generator of S_3 . Right differential linearity proves surjectivity. In particular, this closed element is not a boundary in P_3 . Otherwise its defect would vanish, since $\beta_3b = 0$.

The long exact sequence has the consecutive terms

$$0 \rightarrow H^{-3}(\tilde{P}_3) \rightarrow H_D^{-3} \rightarrow M_3 \rightarrow H^{-2}(\tilde{P}_3) \rightarrow H_D^{-2} \rightarrow S_3 \rightarrow H^{-1}(\tilde{P}_3) \rightarrow H_D^{-1} \rightarrow 0.$$

Lemma 1.7.1 gives surjectivity of δ_{-3} . Use both surjections in this sequence. All other degrees follow because N_3 has only the two stated components. $\square$

These formulas separate two changes. The translation construction replaces $H^r(O_3)$ by the exact tensor product in (1.6). The native collision terms then impose (1.21). The vanishing $M_3/K = 0$ uses the vacuum tangent relation and Weyl commutators. A quotient taken only over regular functions does not permit this argument. The coefficient integral table specifies δ_{-2} on every cycle, without requiring a coefficient-formality assertion about C_3 . Both connecting maps are nonzero, so the projection is not a quasi-isomorphism.

1.8 THE FIRST FOUR-CURRENT COMPATIBILITY

The binary added module has vacuum output. Tensoring it with a single current gives the ternary pair module through the corresponding closed direct image. Its differential remains zero by the vacuum calculation above. Tensoring with two currents introduces a new collision.

In E_4 , for $i < j$ and the remaining labels $l < m$, set

$$w_{ij} = k[\mathbf{1}\delta_{ij}\partial_{t_{ij}}|J_l|J_m].$$

For a decomposition into two pairs, order the pair blocks lexicographically and let

$$d_{ij|lm} = k^2[\mathbf{1}\delta_{ij}\partial_{t_{ij}}|\mathbf{1}\delta_{lm}\partial_{t_{lm}}].$$

These have degrees -3 and -2 , respectively. They are supported on their actual pair and double-pair diagonals. The double-pair class is nonzero because both transfer derivatives and k^2 are nonzero. There is no outer pole between the two vacuum blocks.

PROPOSITION 1.8.1. The six pair terms cannot all be assigned zero differential while retaining their native images. The smallest right differential subcomplex containing them includes the three double-pair modules. Its differentials on the displayed generators are

w	w_{12}	w_{34}	w_{13}	w_{24}	w_{14}	w_{23}	(1.22)
Qw	$-d_{12 34}$	$d_{12 34}$	$-d_{13 24}$	$d_{13 24}$	$-d_{14 23}$	$d_{14 23}$	

Every double-pair term is closed. The resulting correction satisfies the four-current equation $Q^2[J_1|J_2|J_3|J_4] = 0$.

Proof. In $[\mathbf{1}_{ij}|J_l|J_m]$, both collisions involving the vacuum have zero principal part. The remaining term in (1.9) is $[\mu(J_l, J_m)|\mathbf{1}_{ij}]$ with positive sign. Putting its blocks in the prescribed order gives precisely (1.22). Right derivatives preserve these equations. Multiplying the two normal coordinates lowers the two derivatives to the simple double-delta generator. Its right differential closure is the full regular

double-pair vacuum module. Thus these three modules are forced, and they suffice to close the displayed current subspace under Q . Their next differential vanishes because both states are vacuum with regular coefficients.

For four even inputs, the first collision of labels $i < j$ has sign $(-1)^{i+j-3}$ when its block is placed first. Hence

$$Q[J_1|J_2|J_3|J_4] = w_{12} - w_{13} + w_{14} + w_{23} - w_{24} + w_{34}.$$

Each complementary pair has the same first sign and opposite second signs in (1.22). The three double-pair coefficients cancel individually. $\square$

This calculation specifies the next differential on the regular four-current collision subspace. The binary and ternary maps above have their full stated global coefficient domains. A map on every four-input ordered coefficient requires additional operations on the other four-point partition summands. An all-arity bar coalgebra morphism also requires ordered-surjection and deconcatenation identities. Those identities are not consequences of the individual cochain maps constructed here.

FOUR-INPUT COEFFICIENTS AND COLLISION MODULES

2.1 THE FIVE WORDS OF AN ORDERED SECTOR

Retain the nonzero scalar k , the unextended states V_k , and the principal-part normalization of Chapter 1. Apply the definition of C_n there with $n = 4$. Thus $R = \mathbb{C}[z_1, z_2, z_3, z_4]$, and the coefficient families range over faces of the simplex on twenty-four orders. Let $D = D_4$ be its Weyl algebra. All coefficients and forms are algebraic, and all state and differential-operator sums are finite.

Fix an order $\sigma = i_1 i_2 i_3 i_4$. For this sector write $a_r = a_{i_r}$, $t_r = z_{i_r} - z_{i_{r+1}}$, $q_r = q_{i_r i_{r+1}}$, and $\omega_r = dq_r/t_r$. The untranslated words are

$$\begin{aligned} e &= [a_1 | a_2 | a_3 | a_4], \\ m_1 &= [b_{i_1 i_2} | a_3 | a_4], \\ m_2 &= [a_1 | b_{i_2 i_3} | a_4], \\ m_3 &= [a_1 | a_2 | b_{i_3 i_4}], \\ d &= [b_{i_1 i_2} | b_{i_3 i_4}]. \end{aligned}$$

All five words have degree -4 . The direct sum of their free C_4 -modules over all orders is O_4 . It has one hundred twenty untranslated generators. The bar signs give

$$\begin{aligned} be &= -\omega_1 m_1 - \omega_2 m_2 - \omega_3 m_3, \\ bm_1 &= -\omega_3 d, \quad bm_3 = -\omega_1 d, \quad bm_2 = bd = 0, \\ b(cv) &= d_C c v + (-1)^{|c|} c bv. \end{aligned} \tag{2.1}$$

For example, the two contributions to $b^2 e$ are $-\omega_1 \omega_3 d$ and $-\omega_3 \omega_1 d$, which cancel. Closedness of the ω_r proves $b^2 = 0$ on every word.

Use the translated products (1.2) for all four labels. Their proof of associativity and translation covariance applies without change to four labels. The resulting bar sector P_4 has words v_α for $\alpha \in \mathbb{N}^4$. Its complete translation and differential formulas are

$$L_i(cv_\alpha) = (\partial_i c)v_\alpha + cv_{\alpha+e_i}, \quad w\partial_i = -L_i w, \quad bv_\alpha = L^\alpha bv_0.$$

The coefficient rule in (2.1) is retained. The triangular proof of Proposition 1.2.1 gives

$$P_4 \cong O_4 \otimes_R D, \quad H^r(P_4) = H^r(O_4) \otimes_R D. \tag{2.2}$$

In particular, the original bar embeds as its untranslated subcomplex.

2.2 THE SQUARE OF DISJOINT ADJACENT COLLISIONS

Let σ_{ab} be the order obtained by independently reversing the first pair when $a = 0$ and the last pair when $b = 0$. Thus $\sigma_{11} = \sigma$. The four vertices span a face whose coefficient ring is $R[t_1^{-1}, t_3^{-1}]$. Map the polynomial square with coordinates u, v to this face by

$$u_{\sigma_{00}} = (1 - u)(1 - v), \quad u_{\sigma_{10}} = u(1 - v), \quad u_{\sigma_{01}} = (1 - u)v, \quad u_{\sigma_{11}} = uv. \quad (2.3)$$

The pullback sends q_1 to u , q_3 to v , and q_2 to one. Consequently ω_2 restricts to zero. On a side where u is fixed, the coefficients have no t_1 pole. On a side where v is fixed, they have no t_3 pole. These assertions follow from the actual face coefficient rings, including their endpoints.

For a one-form c and a two-form c_2 , write

$$c|_{\square} = g(u, v)du + h(u, v)dv, \quad c_2|_{\square} = G(u, v)du \wedge dv.$$

Then

$$(d_C c)|_{\square} = (\partial_u h - \partial_v g)du \wedge dv.$$

The edge of the first pair is $v = 1$. The edge of the last pair is $u = 1$. The edge of the middle pair remains the ordinary edge $\{\sigma, \sigma(23)\}$. Every edge coordinate equals one at σ .

Put $p_r = p(t_r) = Y(J, t_r)J$, with the states based at the second point of their pair. For a coefficient H on the square define

$$\begin{aligned} W_{00}(H) &= \text{pp}_{t_1} \text{pp}_{t_3} \left(p_1 p_3 \int_0^1 \int_0^1 uv H \, du \, dv \right), \\ W_{10}(H) &= \text{pp}_{t_1} \text{pp}_{t_3} \left(t_1 p_1 p_3 \int_0^1 \int_0^1 v H \, du \, dv \right), \\ W_{01}(H) &= \text{pp}_{t_1} \text{pp}_{t_3} \left(p_1 t_3 p_3 \int_0^1 \int_0^1 u H \, du \, dv \right), \\ W_{11}(H) &= \text{pp}_{t_1} \text{pp}_{t_3} \left(t_1 p_1 t_3 p_3 \int_0^1 \int_0^1 H \, du \, dv \right). \end{aligned}$$

The multiplication and both integrations precede the two principal parts. These outputs lie on the double-pair diagonal, in block order $i_1 i_2 | i_3 i_4$. This notation uses the exterior tensor product of the two oriented binary transfer maps. For $\sigma = 1234$, its density convention is

$$dz_1 \wedge dz_2 \wedge dz_3 \wedge dz_4 = dt_1 \wedge dy_1 \wedge dt_3 \wedge dy_3 = -dt_1 \wedge dt_3 \wedge dy_1 \wedge dy_3, \quad y_1 = z_2, \quad y_3 = z_4. \quad (2.4)$$

Normal derivative changes use this equality and the Weyl action. No normal derivative is discarded.

2.3 THE GRADED COMPARISON ON EVERY COEFFICIENT

Let E_4 be (1.8) for four labels. Its summands are the separated locus, six pair diagonals, four triple diagonals, three double-pair diagonals, and the small diagonal. They have degrees $-4, -3, -2, -2, -1$, respectively. On even blocks the first collision of positions $r < s$ has coefficient $(-1)^{r+s-3}$ when its output block is put first. Subsequent block permutations use the suspended Koszul signs. The normalizations are (1.7) and (2.4).

For an edge one-form $a(u)du$, use v_a and q_a from (1.10). Let $\iota_r(w)$ mean insertion of the merged block w in position r , retaining the other two currents in their original order. For instance, $\iota_2(w) =$

$[J_{i_1}|w|J_{i_4}]$. If $c \in C_4^1$, denote its actual three edge coefficients by a_r . Define the degree-zero R -linear map $B : O_4 \rightarrow E_4$ by

	$f \in C_4^0$	$c \in C_4^1$	$c_2 \in C_4^2$
e	$f(\sigma)[J_{i_1} J_{i_2} J_{i_3} J_{i_4}]$	$\sum_{r=1}^3 (-1)^{r-1} \iota_r(v_{a_r})$	$W_{\text{oo}}(G)$
m_1	$\circ$	$-\iota_1(q_{a_1})$	$-W_{\text{io}}(G)$
m_2	$\circ$	$+\iota_2(q_{a_2})$	$\circ$
m_3	$\circ$	$-\iota_3(q_{a_3})$	$-W_{\text{oi}}(G)$
d	$\circ$	$\circ$	$W_{\text{ii}}(G)$.

(2.5)

All higher coefficient degrees have zero image. The table includes every untranslated source summand. Extend it to P_4 using (1.13). The recursion is unique by (2.2). It sends every unmerged zero-form jet to its actual translated Heisenberg insertion.

2.4 THE COMPLETE BOUNDARY IDENTITIES

All formulas below use one ordered sector. Let $T_r(w)$ be the triple-diagonal part of $Q\iota_r(w)$. Let $C(w)$ be the double-pair output $[\mu(J_{i_1}, J_{i_4})|w]$ with the middle pair in its second block. The coefficient of w is retained during the spectator collision. The corresponding transfer and coordinate permutation are part of this definition. The unshuffle rule gives

$$(Q\iota_2(w))_{\text{double}} = -C(w).$$

For square coefficients put

$$\Delta_u h = h(\mathbf{1}, v) - h(\circ, v), \quad \Delta_v g = g(u, \mathbf{1}) - g(u, \circ),$$

and define the double-pair terms

$$\begin{aligned} D_1(c) &= \text{pp}_{t_1} \text{pp}_{t_3} \left(t_1 p_1 p_3 \int_{\circ}^{\mathbf{1}} v \Delta_u h \, dv \right), \\ D_3(c) &= -\text{pp}_{t_1} \text{pp}_{t_3} \left(p_1 t_3 p_3 \int_{\circ}^{\mathbf{1}} u \Delta_v g \, du \right), \\ D_d(c) &= \text{pp}_{t_1} \text{pp}_{t_3} \left(t_1 p_1 t_3 p_3 \left(\int_{\circ}^{\mathbf{1}} \Delta_v g \, du - \int_{\circ}^{\mathbf{1}} \Delta_u h \, dv \right) \right). \end{aligned} \tag{2.6}$$

In D_1 , the factor $t_1 p_1$ is multiplied by a coefficient regular in t_1 . It therefore has only vacuum principal part. The analogous statement holds for the third pair in D_3 . Each summand of D_d has a vacuum block for the same reason.

Let $\beta = QB - Bb$. Let Q_{na} retain the three nonadjacent pair collisions of the four-current insertion. For 1234 it retains the terms $-w_{13} + w_{14} - w_{24}$ of Proposition 1.8.1. Let $\Delta_r f = f(\sigma) - f(\sigma(r, r + \mathbf{1}))$.

PROPOSITION 2.4.1. On coefficient degree zero the defect is

$$\begin{aligned} \beta(fe) &= f(\sigma) Q_{\text{na}}[J_{i_1}|J_{i_2}|J_{i_3}|J_{i_4}], \\ \beta(fm_r) &= (-1)^{r-1} k \iota_r \left(\text{pp}_{t_r} \frac{\Delta_r f}{t_r} \mathbf{1} \right), \quad r = 1, 2, 3, \\ \beta(fd) &= \circ. \end{aligned} \tag{2.7}$$

On coefficient degree one it is

$$\begin{aligned}\beta(ce) &= \sum_{r=1}^3 (-1)^{r-1} T_r(v_{a_r}) + C(v_{a_2}), \\ \beta(cm_1) &= -T_1(q_{a_1}) + D_1(c), \\ \beta(cm_2) &= +T_2(q_{a_2}) - C(q_{a_2}), \\ \beta(cm_3) &= -T_3(q_{a_3}) + D_3(c), \\ \beta(cd) &= D_d(c).\end{aligned}$$

On coefficient degree two, $\beta(c_2v) = QB(c_2v)$ for every seed v . On every higher coefficient degree it vanishes. These identities extend to all input jets by right differential linearity.

Proof. For fe , each adjacent channel cancels by (1.11). The three nonadjacent channels remain with their unshuffle signs. For fm_r , the only nonzero image of its source differential is the one-form image of dcf . The coefficient $f\omega_s d$ has degree one and zero image. The endpoint identity gives the displayed vacuum output. Both terms vanish for fd .

For coefficient degree one, all triple outputs come from QB . The square images lie on the double-pair diagonal. It remains to compute the latter components of $QB - Bb$. Write $H = \partial_u h - \partial_v g$. The coefficient of $p_1 p_3$ in the double-pair part of $Bb(ce)$ is

$$\int \int uvH + \int \int vh - \int \int ug = \int_0^1 vh(\mathbf{1}, v) dv - \int_0^1 ug(u, \mathbf{1}) du.$$

The first two signs use $c\omega_1 = -h du \wedge dv/t_1$ and $c\omega_3 = g du \wedge dv/t_3$. These boundary terms equal the spectator collisions of $\iota_3(v_{a_3})$ and $\iota_1(v_{a_1})$, with signs $+$ and $-$. The middle-pair spectator collision has no square face and is $+C(v_{a_2})$.

For cm_1 , the coefficient of $t_1 p_1 p_3$ in Bb is

$$- \int \int vH + \int \int g = \int_0^1 g(u, \mathbf{1}) du - \int_0^1 v\Delta_u h dv.$$

The first term is its spectator collision in QB . Subtracting leaves $D_1(c)$. For cm_3 , the coefficient of $p_1 t_3 p_3$ is

$$- \int \int uH - \int \int h = - \int_0^1 h(\mathbf{1}, v) dv + \int_0^1 u\Delta_v g du.$$

Subtract it from the spectator collision to obtain $D_3(c)$. The middle mixed word has no nonzero symbol differential and no square image. Thus its entire defect is $Q\iota_2(q_{a_2})$. Finally, $Bd_C(cd) = W_{\Pi}(H)$. Ordinary integration of H gives $D_d(c)$ with the displayed sign.

For coefficient degree two, every term in Bb has coefficient degree at least three, hence zero image. The same degree argument proves the final vanishing assertion. All integrations, principal parts, and coordinate changes commute with the right differential action through the prescribed jet recursion. $\square$

2.5 A COMPLEX OF REGULAR COLLISION STATES

Let $U = \text{span}\{T^r J : r \geq 0\}$ and let $U^{(2)}$ be the span of products of two such states in V_k . Thus $U^{(2)} = \text{Sym}^2(U)$ with the total translation. These spaces have no state differential. Identify $U^{(2)}$ with

symmetric polynomials in two variables λ, μ by polarized monomials. Total translation is multiplication by $\lambda + \mu$. The invariant ring is $\mathbb{C}[\lambda, \mu]^{S_2} = \mathbb{C}[\lambda + \mu, (\lambda - \mu)^2]$. Consequently

$$U^{(2)} = \bigoplus_{a \geq 0} \mathbb{C}[T] h_a, \quad h_a = \sum_{r=0}^{2a} (-1)^r \binom{2a}{r} (T^r J)(T^{2a-r} J). \quad (2.8)$$

In particular $h_0 = J^2$. The invariant-ring equality proves independence and completeness of these translation generators.

The notation below means regular position coefficients and their full right differential closure inside E_4 . It does not add outer inverses. Let $\mathcal{A}$ be the sum over six pair partitions of vacuum on the pair and U on each remaining singleton. Let V be the sum over three double-pair partitions of vacuum on both blocks. Let T_3 be the sum over four triple partitions of U on the triple and U on the singleton. Let W be the sum over double-pair partitions of $(\mathbb{C}\mathbf{1} \boxtimes U^{(2)}) \oplus (U^{(2)} \boxtimes \mathbb{C}\mathbf{1})$. Let S_0 and S_2 be vacuum and $U^{(2)}$, respectively, on the small diagonal. Each closed direct image retains every normal derivative.

These are explicit Weyl modules. For example, the pair summand of $\mathcal{A}$ is

$$D/(t_{ij}D + (\partial_i + \partial_j)D),$$

generated by $[\mathbf{1}\delta_{ij}|J_l|J_m]$. The triple summand with block $B = \{i, j, k\}$ is $D/(t_{ij}D + t_{ik}D)$. Its two surviving current translations are independent. For a double-pair partition $ij|lm$, the vacuum summand is

$$D/(t_{ij}D + t_{lm}D + (\partial_i + \partial_j)D + (\partial_l + \partial_m)D).$$

Each vacuum-first summand of W has the presentation

$$\bigoplus_{a \geq 0} D/(t_{ij}D + t_{lm}D + (\partial_i + \partial_j)D),$$

with generators $\mathbf{1} \boxtimes h_a$. Exchange the pairs for the other summand. Put $P = \partial_1 + \partial_2 + \partial_3 + \partial_4$ and $L = t_{14}D + t_{24}D + t_{34}D$. Then $S_0 = D/(L + PD)$ and $S_2 = \bigoplus_{a \geq 0} D/L$. Normal ordering and (2.8) prove that all these presentations have exactly the stated relations.

The only nonzero differentials among these modules are

$$N^{-3} = \mathcal{A} \xrightarrow{d_A} N^{-2} = V \oplus T_3 \oplus W \xrightarrow{d_T} N^{-1} = S_0 \oplus S_2.$$

Here d_A lands in V , and d_T is zero except on T_3 , where it lands in S_0 . Both maps are the native current collision, including transfers. For arbitrary translated currents its scalar coefficient is

$$K_{r,s}(x) = k(-1)^r (r+s+1)! x^{-r-s-2}, \quad \text{pp}_x Y(T^r J, x) T^s J = \text{pp}_x K_{r,s}(x) \mathbf{1}. \quad (2.9)$$

The signs for d_A are precisely (1.22), also for simple inner deltas. For d_T , the triple block is first, so its binary collision sign is positive. If $B = \{b_1 < b_2 < b_3\}$ and the remaining label is l , use staged coordinates

$$z_{b_1} - z_{b_3}, \quad z_{b_2} - z_{b_3}, \quad z_{b_3} - z_l, \quad z_l.$$

Their density sign relative to $dz_1 \wedge \cdots \wedge dz_4$ is $(-1)^{4-l}$. This fixes every component of d_T in the common small-diagonal normalization. The inner normal derivatives are transported by the displayed linear coordinate change.

LEMMA 2.5.1. The complex N is a subcomplex of E_4 , and $\beta(P_4) \subset N$.

Proof. In a pair summand of A , collisions involving the vacuum have zero principal part. The remaining two currents collide by (2.9), giving V . The two currents of a triple summand give S_0 . Every output in V or W has a vacuum block and regular outer coefficients, so its next collision is zero. This proves the subcomplex assertion, including all right derivatives.

The zero-form defects in (2.7) lie in A . For degree one, a singular collision of a quadratic current state and one current leaves one current. It therefore gives T_3 with the remaining spectator. A spectator collision leaves a vacuum block and a quadratic or vacuum block, giving $W \oplus V$. The side defects in (2.6) have the same property by side regularity.

For degree two, the only images before applying Q are two quadratic or vacuum blocks on a double-pair diagonal. Their regular outer collision leaves either two currents or vacuum. For completeness, if $a = (T^r J)(T^s J)$ and $b = (T^u J)(T^v J)$, its singular part is the principal part of

$$\begin{aligned} & K_{ru}(x)(e^{xT} T^s J)(T^v J) + K_{rv}(x)(e^{xT} T^s J)(T^u J) \\ & + K_{su}(x)(e^{xT} T^r J)(T^v J) + K_{sv}(x)(e^{xT} T^r J)(T^u J) \\ & + K_{ru}(x)K_{sv}(x)\mathbf{I} + K_{rv}(x)K_{su}(x)\mathbf{I}. \end{aligned}$$

This is the finite Wick formula with one or two contractions. It gives $S_2 \oplus S_0$ after every regular outer coefficient and finite inner normal derivative. All higher coefficient defects vanish by Proposition 2.4.1. $\square$

2.6 THE NATIVE FOUR-INPUT COCHAIN MAP

THEOREM 2.6.1. On all of X^4 , the formulas

$$\tilde{P}_4 = P_4 \oplus N, \quad \tilde{b}(x, n) = (bx, \beta x + Qn), \quad F_4(x, n) = Bx + n \quad (2.10)$$

define a complex of right differential modules and a cochain map $F_4 : \tilde{P}_4 \rightarrow E_4$. They include every ordered word, every global compatible-family coefficient, every finite input translation, and all fifteen target partition types. The construction is equivariant for every permutation of the four labels.

Proof. Expanding $\beta = QB - Bb$ gives $Q\beta + \beta b = 0$. Lemma 2.5.1 makes this an identity in N . Thus

$$\tilde{b}^2(x, n) = (0, \beta bx + Q\beta x + Q^2 n) = 0.$$

Direct substitution also gives $QF_4 - F_4\tilde{b} = QB - Bb - \beta = 0$. The complete component identities are Proposition 2.4.1. They prove that every entry in the source differential has its specified native support and finite normal expansion.

A permutation carries each actual edge and square to the edge and square of the relabeled word. It preserves the word-relative coordinates and their vertex value one. Evaluation, polynomial integration, the Heisenberg operation, and right transfer commute with this relabeling. The coefficient wedge signs and the suspended block signs are both retained. For example, exchanging the two disjoint pairs reverses $du \wedge dv$ and exchanges two degree- -1 collision blocks. The two minus signs agree in the square formula. All other signs are the unshuffle and tensor signs used in (2.5). Therefore B , its defect, N , and (2.10) are permutation-equivariant. $\square$

The source differential includes the triple and small-diagonal terms of the defect table. Its pair terms also have the nonzero next differential d_A . Declaring all new collision terms closed would contradict (1.22). The level and the original double-pole coefficients remain unchanged.

2.7 THE COHOMOLOGY OF THE ADDED COMPLEX

For a pair ij , let D_{ij} be the two-coordinate Weyl algebra and put

$$I_{ij} = t_{ij}^2 D_{ij} + (t_{ij} \partial_i - 1) D_{ij} + (\partial_i + \partial_j) D_{ij}.$$

The map $D_{ij} \rightarrow$ vacuum pair module, sending its generator to $k \delta_{ij} \partial_{t_{ij}}$, is surjective and has kernel I_{ij} . Normal ordering proves this presentation: the normal relations reduce to the simple delta and its successive derivatives, and the tangent derivative kills vacuum. For disjoint pairs, regard $I_{ij} I_{lm}$ as the external tensor product right ideal inside $D = D_{ij} \otimes D_{lm}$.

To describe d_T algebraically, set $L_B = t_{b_1 b_3} D + t_{b_2 b_3} D$ and $P_B = \sum_{i \in B} \partial_i$. Scale the external triple generator by the density sign $(-1)^{4-l}$ just specified. Then d_T sends its class to $k \delta_{1234} P_B$. Thus define the explicit Weyl-module kernel

$$\mathcal{K}_T = \left\{ (A_B \bmod L_B)_B \in \bigoplus_{|B|=3} D/L_B : \sum_B P_B A_B \in L + PD \right\}. \quad (2.11)$$

The condition is well-defined because each P_B commutes with the internal normal coordinates of its block.

THEOREM 2.7.1. The maps $d_A : A \rightarrow V$ and $d_T : T_3 \rightarrow S_0$ are surjective. The cohomology of N is

$$\begin{aligned} H^{-3}(N) &\cong \bigoplus_{ij|lm} D/(I_{ij} I_{lm}), \\ H^{-2}(N) &\cong \mathcal{K}_T \oplus W, \\ H^{-1}(N) &\cong S_2, \quad H^r(N) = 0 \quad (r \neq -3, -2, -1). \end{aligned} \quad (2.12)$$

The three generators in the first line map to $w_{ij} + w_{lm}$, using lexicographically ordered complementary pairs. Their normalizations are those of Proposition 1.8.1.

Proof. In each complementary pair channel, d_A applied to a simple inner delta and two currents gives a nonzero multiple of the remaining double pole. Multiplication by the remaining normal coordinate gives the simple double-pair vacuum generator. Right differential closure proves surjectivity. The same argument with a triple delta and two current blocks proves surjectivity of d_T .

For each complementary pair channel, tensor the two surjections $D_{ij} \rightarrow D_{ij}/I_{ij}$ and $D_{lm} \rightarrow D_{lm}/I_{lm}$ over $\mathbb{C}$. The resulting three-term tensor complex is exact in its middle term. Its first map sends the four-current generator to $w_{ij} + w_{lm}$, and its second map is the corresponding component of d_A . The kernel of its first map is

$$(I_{ij} \otimes D_{lm}) \cap (D_{ij} \otimes I_{lm}) = I_{ij} \otimes I_{lm} = I_{ij} I_{lm}.$$

Choose vector-space complements to the two ideals to verify the intersection equality directly. There are three independent complementary channels, proving the first line of (2.12).

The image of d_A is all of V and has no component in $T_3 \oplus W$. The kernel of the next differential there is $\mathcal{K}_T \oplus W$ by (2.11). Finally, d_T removes exactly S_0 from $S_0 \oplus S_2$. This proves every remaining assertion. $\square$

2.8 RELATIVE COHOMOLOGY OF THE FOUR-INPUT SOURCE

Write $H_D^r = H^r(O_4) \otimes_R D$ and let $\pi : \tilde{P}_4 \rightarrow P_4$ be the projection. The exact sequence $0 \rightarrow N \rightarrow \tilde{P}_4 \rightarrow P_4 \rightarrow 0$ gives connecting maps

$$\delta_r : H_D^r \longrightarrow H^{r+1}(N), \quad \delta_r[x] = [\beta x].$$

The defect table specifies these maps on every coefficient cycle and input jet. In the convention $\text{Cone}(\pi)^r = P_4^r \oplus \tilde{P}_4^{r+1}$, the inclusion of $N[1]$ is a quasi-isomorphism onto this cone. Indeed, the quotient is the contractible cone of the identity on P_4 . Thus the relative groups are exactly

$$\begin{aligned} H^{-4}(\text{Cone } \pi) &= \bigoplus_{ij|lm} D/(I_{ij}I_{lm}), \\ H^{-3}(\text{Cone } \pi) &= \mathcal{K}_T \oplus W, \\ H^{-2}(\text{Cone } \pi) &= S_2, \end{aligned}$$

and they vanish in every other degree. In particular, π is not a quasi-isomorphism.

The ordinary cohomology change has the exact form

$$0 \longrightarrow \text{coker } \delta_{r-1} \longrightarrow H^r(\tilde{P}_4) \longrightarrow \ker \delta_r \longrightarrow 0, \quad (2.13)$$

where $\delta_r = 0$ unless $r = -4, -3, -2$. Here a cokernel has target $H^r(N)$, and a kernel has source H_D^r . These conventions specify (2.13) also in degrees outside that range. No splitting is asserted.

PROPOSITION 2.8.1. The connecting map $\delta_{-2} : H_D^{-2} \rightarrow S_2$ is surjective. Consequently $H^{-1}(\tilde{P}_4) \rightarrow H_D^{-1}$ is an isomorphism.

Proof. For $r \geq 0$, the coefficient

$$\gamma_r = \frac{dq_{12} \wedge dq_{34}}{t_{12}^{r+2} t_{34}^2}$$

is an actual closed global two-form. On every face lacking a required inverse, the corresponding numerator differential vanishes. Since $bd_{1234} = 0$, the element $\gamma_r d_{1234}$ is a cycle of degree -2 . Its double-pair image is the product of the two principal parts

$$\text{PP}_{t_1} \frac{p(t_1)}{t_1^{r+1}}, \quad \text{PP}_{t_3} \frac{p(t_3)}{t_3}.$$

The quadratic part of the first factor is

$$\sum_{a=0}^r \frac{(T^a J)J}{a!} \frac{\delta_{12} \partial_{t_1}^{r-a}}{(r-a)!},$$

and the second quadratic factor is $J^2 \delta_{34}$. The vacuum terms have zero outer collision.

Filter $U^{(2)}$ by the submodules generated by $h_0, \dots, h_m$ under T . This filtration is preserved by every right differential operator after direct image. Take $r = 2m$ and suppose the preceding filtration step already lies in the connecting image. All terms with $a < r$ belong to that preceding step. For the remaining quadratic state $(T^r J)J$, the coefficient of the outer double pole against J^2 is

$$2k(r+2)(T^r J)J.$$

This follows from the four single contractions in the displayed Wick formula. The coefficients of all higher outer poles lie in the preceding filtration step. Multiplication by the outer difference $v = y_1 - y_3$ removes its simple-pole term and turns the double pole into a simple pole. Modulo the preceding step, the resulting coefficient in the staged double-pair orientation is

$$\frac{2k(r+2)}{r! 2^r} h_m \delta_{1234}.$$

Here $(T^{2m}J)J \equiv 2^{-2m}h_m$ modulo the preceding step, by the symmetric-polynomial description. The displayed scalar is nonzero. For $m = 0$ this starts with $4kJ^2\delta_{1234}$. Induction and right differential closure therefore generate all of S_2 . The vacuum small-diagonal terms vanish in $H^{-1}(N)$ by Theorem 2.7.1. This proves surjectivity and the stated consequence of the long exact sequence. $\square$

The lowest source group and its connecting map are more explicit. For a fixed order put $r_j = q_j/t_j$ in the full position localization and set

$$\begin{aligned} e' &= e + r_1 m_1 + r_2 m_2 + r_3 m_3 + r_1 r_3 d, \\ m'_1 &= m_1 + r_3 d, & m'_3 &= m_3 + r_1 d. \end{aligned}$$

These symbols are closed there. Their following multiples belong to the actual global coefficient source:

$$z_\sigma = t_1 t_2 t_3 e', \quad u_\sigma = t_3 m'_1, \quad m_{2,\sigma}, \quad v_\sigma = t_1 m'_3, \quad d_\sigma. \quad (2.14)$$

PROPOSITION 2.8.2. The one hundred twenty classes in (2.14) form a free R -basis of $H^{-4}(O_4)$. If $c_4 = Q[J_1|J_2|J_3|J_4]$ and $p_\sigma = t_1 t_2 t_3$, then

$$\delta_{-4}(z_\sigma) = \text{sgn}(\sigma) p_\sigma c_4, \quad (2.15)$$

and the connecting map kills the other four basis elements of each order. In the three cyclic summands of $H^{-3}(N)$, the coefficient vector of c_4 is $(1, -1, 1)$.

Proof. The triangular change from e, m_1, m_2, m_3, d to e', m'_1, m'_2, m'_3, d makes the differential the coefficient differential on the full simplex. A closed lowest element therefore has constant coefficients in this basis. At the reverse-order vertex all three q_j vanish. Its regular coefficients force these constants to belong to R .

The three actual adjacent edges force $t_j \mid f$ for the coefficient f of e' . The three distinct linear factors are relatively prime, hence $t_1 t_2 t_3 \mid f$. Subtract the corresponding multiple of z_σ . For the remaining element, the four corners of (2.3) force t_3 to divide the coefficient of m'_1 and t_1 to divide that of m'_3 . Subtract their displayed multiples. The remaining coefficients of m_2 and d are arbitrary elements of R . This proves spanning, and the triangular leading coefficients prove independence. There is no source degree below -4 , hence no incoming boundary.

All untranslated mixed zero-form images vanish under B . Thus $Bz_\sigma = p_\sigma [J_{i_1}|J_{i_2}|J_{i_3}|J_{i_4}]$. Applying Q proves (2.15). The final signs follow from the complementary pair calculation in Proposition 1.8.1. $\square$

For an explicit coefficient description, put $x = z_1 - z_4$, $y = z_2 - z_4$, and $z = z_3 - z_4$. The twenty-four path polynomials p_σ generate

$$(x, y, z)(x(y - z), y(x - z)).$$

Their cubic span has dimension six. Here is an explicit verification of the ideal equality. Put $A = x(y - z)$ and $B = y(x - z)$. Expansion gives

$$\begin{aligned} zB &= -p_{1342}, & yB &= -p_{1342} - p_{1324}, \\ yA &= p_{1243} - p_{1342} - p_{1324}, & zA &= p_{1234} + p_{1243} - p_{1342}, \\ xA &= p_{1234} + p_{1243} - p_{1342} - p_{2314}, & xB &= -p_{2314} - p_{1342} - p_{2134}. \end{aligned}$$

The six cubics xA, xB, yA, yB, zA, zB are independent. Indeed, a dependence is a relation between A and B with linear coefficients. Their relatively prime quadratic factors force both coefficients to vanish. Every path cubic vanishes on the three coordinate axes and on $x = y = z$. These four independent conditions leave a six-dimensional space of cubics. The six displayed generators lie in that space, so it is exactly their span. This proves the asserted ideal equality. Consequently the image in (2.15) is given by this explicit polynomial ideal acting on the cyclic vector $(1, -1, 1)$ through right multiplication, followed by right differential closure. The presentation retains all Weyl relations and does not identify that image with the whole target.

2.9 ORDERED THREE-PLUS-ONE CUTS

Fix a word sector $\sigma = i_1 i_2 i_3 i_4$ and its prefix $I = \{i_1, i_2, i_3\}$. Let $l = i_4$. On coefficients, restrict to the face of orders ending in l . It is the actual three-input coefficient family on I , with the additional position z_l . On bar words, retain a cut only when it falls between blocks. Thus $e \mapsto e_{i_1 i_2 i_3} \otimes a_l$ and m_1, m_2 give their ternary mixed words tensored with a_l . The words m_3, d have no cut of this type and map to zero. The coefficient differential and every surviving multiplication face agree with the tensor bar differential. Hence this is an ordered cochain cut on $P_{4,\sigma}$.

Let P_I be the current translation module generated by a_l , of degree -1 , with $a_l \mapsto J_l$ and zero differential. It maps isomorphically onto the regular translated-current submodule of E_l . Let

$$U_{I|l} = X^4 \setminus \bigcup_{i \in I} \Delta_{il}.$$

On this open set, all crossing partition summands vanish. The native remaining complex is the exterior tensor product of E_3 and E_1 , with its cross coefficients localized. This is the explicit carrier factorization in [1], equation (3.4.II.5).

PROPOSITION 2.9.1. For the fixed word sector and this ordered cut, the comparison F_4 agrees on $U_{I|l}$ with $F_3 \boxtimes F_1$. The source cut extends from $P_{4,\sigma}$ to $P_{4,\sigma} \oplus N$ by

$$A_{ij} \longrightarrow M_{ij} \boxtimes P_1 \quad (ij \subset I), \quad (T_3)_I \longrightarrow S_3 \boxtimes P_1,$$

and zero on the other partition summands after restriction to this open set. These maps retain each input translation and normal derivative. The analogous assertion holds for the ordered one-plus-three cut.

Proof. The noncrossing edge formulas in (2.5) are exactly (1.14) with the last suspended current. Every square image involves the pair $i_3 i_4$, so it vanishes on $U_{I|l}$. The crossing mixed images vanish for the same reason. This proves equality of the graded comparisons on all coefficients and jets.

The indicated maps on N use the identity on the collision factors and the inverse of $P_1 \rightarrow U$ on the spectator. They are well-defined right differential-module maps. Every nonzero differential from their pair terms crosses the cut, hence vanishes on this open set. The other retained term is already closed. Thus the added-module maps are cochain maps.

Take the defect of the already proved graded equality. The ordered source cut commutes with b , and F_1 has zero defect. The resulting equality is precisely the compatibility of the added source terms with $\beta_3 \boxtimes F_1$. Its native inclusion is injective on these retained added modules, so the equality holds in the extended sources themselves. This proves the asserted cochain comparison. For the one-plus-three cut, moving a coefficient form past the first degree-1 current gives its usual tensor sign. That sign is the edge sign $(-1)^{r-1}$ in (2.5), proving the second assertion. $\square$

This statement concerns the cuts belonging to a fixed ordered word. A single coproduct on all words must also specify its nonprefix components. The ordered deconcatenation makes such a component zero, whereas the native Chevalley coproduct includes its unshuffle term. For example, the $123|4$ component of the ordered cut of e_{4123} is zero. The same component of the native separated image $[J_4|J_1|J_2|J_3]$ is nonzero on the distinct-position locus. Thus these all-word comparisons do not intertwine ordered deconcatenation with the native unshuffle coproduct.

2.10 THE TWO-PLUS-TWO TORSION OBSTRUCTION

Even on the single order 1234 , the added pair modules impose another restriction. Use the binary source $\tilde{P}_2$ and map F_2 from Proposition 1.4.1 on each of the pairs 12 and 34 . Let $v_{12} = k\mathbf{1}\delta_{12}\partial_{t_{12}}$ and let $w_{12} = [v_{12}|J_3|J_4] \in \mathcal{A}$. Its source differential is the complementary double-pair vacuum term. The collision modules used in this calculation have regular cross-position coefficients.

PROPOSITION 2.10.1. There is no strict right differential-module cochain cut

$$P_{4,1234} \oplus N \longrightarrow \tilde{P}_2(12) \boxtimes \tilde{P}_2(34)$$

that preserves the native comparison on w_{12} and on its double-pair differential. The target cut is the partition projection for $12|34$. The same obstruction holds after localizing cross-pair differences.

Proof. The underlying regular-function module of P_2 has no torsion in its pair normal coordinate. Indeed, filter its induced presentation by differential order. Its associated graded is a direct sum of the torsion-free coefficient modules of O_2 . The coefficient modules embed into their full position localization, proving their torsion-freeness. Therefore any map from a module supported on Δ_{12} to $P_2(12)$ is zero. The same statement applies to Δ_{34} and remains true after cross localization.

In degree -3 , the image of w_{12} must consequently lie in $N_2(12) \boxtimes P_2(34)^{-2}$. There is no term $N_2 \boxtimes N_2$ in that degree. In degree -2 , the image of its double-pair differential must lie in $N_2(12) \boxtimes N_2(34)$, by torsion in both normals. The component of the chain equation in $N_2(12) \boxtimes P_2(34)$ therefore forces the second factor of the first image to be a cycle. Extracting each independent first-factor transfer coefficient makes this a statement about individual cycles of $P_2(34)$.

The lowest binary cycle calculation gives $Z^{-2}(P_2) = zD_2 \oplus hD_2$, where $z = te + uh$. Its separated comparison has image exactly

$$B_2(Z^{-2}(P_2)) = [J|J] \cdot tD_2.$$

The separated translated-current module is free cyclic over D_2 , because $T^r J \boxtimes T^s J$ are independent. The right ideal tD_2 is proper: the nonzero simple delta is a quotient generator killed by t . Hence the pure separated current $[J|J]$ does not belong to this cycle image.

Preserving the comparison on w_{12} requires precisely that pure current as the coefficient of v_{12} . The cycle condition just proved makes this impossible. Cross localization does not invert t_{34} and preserves the nonzero delta quotient on its diagonal. Thus the same contradiction holds there. $\square$

BIBLIOGRAPHY

- [1] A. Beilinson and V. Drinfeld, *Chiral Algebras*, American Mathematical Society, 2004, §3.1.1–3.1.2 and §3.4.11, equations (3.4.11.1), (3.4.11.5), pp. 157–158, 182–183.